MIL-E-0022200/8A(SH)
15 October 1980
SUPERSEDING
MIL-E-0022200/8(SHIPS)
7 December 1971
(See 6.4)

MILITARY SPECIFICATION

ELECTRODES, WELDING, COVERED, LOW-HYDROGEN, AND IRON-POWDER LOW-HYDROGEN, CHROMIUM-MOLYBDENUM ALLOY STEEL AND CORROSION RESISTING STEEL

This limited coordination military specification has been prepared by the Naval Sea Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-E-0022200/8(SHIPS). It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used for acquisition purposes.

1. SCOPE

1.1 Scope. This specification covers electrodes for welding alloy steels used for corrosion resistance in certain media, such as petroleum crudes, and for high-temperature strength and resistance to creep.

1.2 Classification. Electrodes shall be furnished in the types, classes, and sizes shown in table I, as specified (see 6.2.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3439

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TABLE I. Types, classes, and sizes.[1/]

Type	Class	Positions of welding	Diameter of core wire (Inch)
MIL-8015-B2 MIL-8016-B2 MIL-8018-B2	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4
MIL-8015-B2L MIL-8018-B2L	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4
MIL-9015-B3 MIL-9016-B3 MIL-9018-B3	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4
MIL-9015-B3L MIL-9018-B3L	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4
MIL-410-15 MIL-410-16	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4
MIL-502-15 MIL-502-16	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4
MIL-502-15-L MIL-502-16-L	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4
MIL-505-15 MIL-505-16	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4

[1/] Type classification of electrodes will be determined according to the capability of the electrode to meet the tests specified herein, when used in the positions of welding specified in table I and in section 4. An electrode shall be approved under ONE type classification; and an electrode shall NOT be approved if it differs in type from that stated in the manufacturer's request for test. The position description indicates the maximum size qualified under this specification. However, reference should be made to the Qualified Products List (QPL) for the selection of electrodes for all positions of welding.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-10430 - Welding Rods and Electrodes; Preparation for Delivery of.
MIL-E-22200 - Electrodes, Welding, Covered; General Specification for.
MIL-S-22698 - Steel Plate, Carbon, Structural, for Ships.

STANDARD

MILITARY

MIL-STD-271 - Nondestructive Testing Requirements for Metals.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSEA 0900-LP-003-9000 - Radiographic Standards for Production and Repair Welds.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 131 - Structural Steel for Ships.
A 370 - Methods and Definitions for Mechanical Testing of Steel Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

B4.0 - Standard Methods for Mechanical Testing of Welds.
Z49.1 - Safety in Welding and Cutting.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

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DEPARTMENT OF LABOR
Code of Federal Regulations, Title 29
Part 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Electrodes furnished under this specification shall conform to the requirements of MIL-E-22200 and as specified herein.

3.2 Electrode composition.

3.2.1 Core wire composition. Core wire composition is optional with the manufacturer.

3.2.2 Covering. The chemical composition of the covering, except as specified in 3.2.2.1 and 3.2.2.2 is optional with the manufacturer.

3.2.2.1 Total iron content. The total iron of the covering of MIL-XX18 type electrodes, including any combined iron in addition to metallic iron powder, shall be not less than 15 percent.

3.2.2.2 Water content. The water content of the covering shall not exceed the percent-by-weight values specified in table II.

TABLE II. Water content.

Type	Percent by weight maximum
MIL-80XX	0.4
MIL-90XX	.2
MIL-410-XX	.6
MIL-502-XX	.6
MIL-505-XX	.6

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3.3 Deposited weld metal.

3.3.1 Mechanical properties. The mechanical properties of weld metal deposited in groove joints shall be as specified in table III. The stress-relieved properties shall be determined for weld metal, postheat treated in accordance with 4.5.1.1 (see table XI).

TABLE III. Mechanical properties.

Type	Ultimate tensile strength		Yield strength[1]		Elongation in 2 inches	
	As-welded	Stress-relieved	As-welded	Stress-relieved	As-welded	Stress-relieved
	lb/in^2 (minimum)	lb/in^2 (minimum)	lb/in^2 (minimum)	lb/in^2 (minimum)	Percent (minimum)	Percent (minimum)
MIL-8015-B2L[2] MIL-9015-B3L MIL-502-15-L MIL-502-16-L	75,000	70,000	50,000	45,000	20 18 17 17	22
MIL-8015-B2[3] MIL-8016-B2	------	80,000	------	67,000	--	19
MIL-8018-B2[3]	------	80,000	------	67,000	--	19
MIL-8018-B2L[2]	80,000	70,000	67,000	60,000	19	19
MIL-9015-B3[3] MIL-9016-B3	------	90,000	------	67,000	--	17
MIL-9018-B3[3]	------	90,000	------	77,000	--	17
MIL-9018-B3L[2]	90,000	80,000	77,000	70,000	17	17
MIL-410-15[3] MIL-410-16 MIL-502-15 MIL-502-16 MIL-505-15 MIL-505-16	------	75,000	------	50,000	--	20

[1] The yield strength shall be determined by the offset method (offset = 0.2 percent).

[2] The as-welded and stress-relieved tensile properties shall be determined for qualification testing of electrodes for the QPL. The as-welded tensile properties shall be determined for comparison inspection, when required, and quality conformance inspection.

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3/ The stress-relieved tensile properties shall be determined for qualification testing of electrodes for the QPL. The stress-relieved tensile properties shall be determined for comparison inspection, when required.

3.4 Chemical composition of deposited weld metal. Chemical composition of deposited weld metal shall be as specified in table IV.

TABLE IV. Chemical composition of deposited weld metal, percent.1/

Type	Carbon	Manganese	Phosphorous	Sulfur	Silicon	Chromium	Molybdenum
MIL-8015-B2 MIL-8016-B2 MIL-8018-B2	0.10	0.90	0.03	0.04	0.60 .60 .80	1.00-1.50	0.40-0.65
MIL-8015-B2L MIL-8018-B2L	.05	.90	.03	.04	1.00 .80	1.00-1.50	.40-0.65
MIL-9015-B3 MIL-9016-B3 MIL-9018-B3	.10	.90	.03	.04	.60 .60 .80	2.00-2.50	.90-1.20
MIL-9015-B3L MIL-9018-B3L	.05	.90	.03	.04	1.00 .80	2.00-2.50	.90-1.20
MIL-410-15 MIL-410-16	.10	1.00	.04	.03	.90	11.00-13.50	----
MIL-502-15 MIL-502-16	.10	1.00	.04	.03	.90	4.00-6.00	.45-0.65
MIL-502-15-L MIL-502-16-L	.05	1.00	.04	.03	.90	4.00-6.00	.45-0.65
MIL-505-15 MIL-505-16	.10	1.00	.04	.03	.60	8.50-10.50	1.00-1.50

1/ Single values are maximum percentages.

3.5 Electrode identification. Electrodes shall be identified as specified in 3.5.10 of MIL-E-22200.

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3.6 Electrode lengths. The specified lengths of end grip electrodes shall be in accordance with table V (see 6.2.1).

TABLE V. Electrode lengths.

Size (Inch)	Length (Inches)
3/32	9, 12 or 14
1/8, 5/32	9 or 14
3/16	14
7/32, 1/4	18

3.7 Alloy identity. The alloy identity shall be in accordance with 3.6.4 of MIL-E-22200.

4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with MIL-E-22200 and as specified herein.

4.2 Qualification tests. Qualification tests shall be in accordance with 4.2 of MIL-E-22200 and tables X and XII herein.

4.3 Comparison inspection. In addition to the examination and tests required in 4.5 of MIL-E-22200, tests specified in table VI shall be conducted as specified in 4.3 of MIL-E-22200, when required by the Naval Sea Systems Command (NAVSEA).

TABLE VI. Comparison inspection.

Electrode type	Sizes	Tests	Requirements
All	All	4.6.8 (MIL-E-22200) Weld test No. 3 (see table X herein)	3.4 and table IV herein 3.3.1 and table III herein

4.4 Quality conformance inspection.

4.4.1 Lot. A lot of electrodes shall be as specified in 4.4.3 and 4.4.6.2 of MIL-E-22200 for inspection level B.

4.4.2 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be in accordance with 4.4.7 through 4.4.8.3 of MIL-E-22200 for tests specified in MIL-E-22200 where required and as specified in 4.4.3 herein for quality conformance weld tests.

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4.4.3 Quality conformance inspection. Quality conformance tests shall be conducted on samples selected as specified in 4.4.2 from electrodes produced from each wet mix covering mixture in each lot as defined in 4.4.3 of MIL-E-22200. One sample shall be subjected to all the tests specified in table VII (i.e., x-ray, mechanical and complete chemical analysis). The samples representing each of the remaining wet mixes of covering mixture in each lot shall be subjected to equivalency tests as specified in table VIII.

TABLE VII. Quality conformance inspection.

Electrode type	Required tests	Procedures
All	Test No. 3 and 8a 1/ 2/	Table X
All	Table XII	Table XII

1/ Types MIL-90XX-B3 and MIL-90XX-B3L electrodes shall be subjected to one rebaking at 800°F ± 25°F for a period of 1/2 hour to 1 hour prior to testing. It should be noted that this rebaking is in addition to the initial manufacturing baking of the electrode covering. This rebaking shall be accomplished except where specifically not required by the acquisition document (see 6.2.1).

2/ Impact properties shall be determined only when specifically required by the acquisition document.

TABLE VIII. Wet mix equivalency tests.

Electrode type	Required tests	Procedures
All All	Test 8a Chemical analysis 1/ 2/ 3/	Table X

1/ Required chemical analysis of the deposited weld metal is for selected elements only as specified in table IX.

2/ Equivalency chemical analysis tests are not required when the lot is as defined in 4.4.3.1(c) of MIL-E-22200.

3/ 4.6.8.1 of MIL-E-22200 may be applied.

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TABLE IX. Wet mix equivalency weld deposit chemical analysis.

Electrode type	Required elements
All	Manganese Silicon Chromium Molybdenum

4.4.4 Unsatisfactory test results. If the results of the first tests are determined to be unsatisfactory, two retests shall be permitted. The results of both retests shall be satisfactory for quality conformance inspection.

4.5 Test procedures.

4.5.1 Weld tests. Table X summarizes the weld tests required under this specification for qualification tests of electrodes and for the inspection specified in tables VI, VII, VIII, and IX.

4.5.1.1 Preheat, interpass, and postheat treatment. The preheat, interpass, and postheat treatment required for the weld tests are summarized in table XI.

4.5.1.2 Weld joint base plate. The weld joint base plate shall be medium steel conformance to MIL-S-22698 or ASTM A 131, with the fusion faces of the groove and backing strip clad with a layer of weld metal. The cladding shall be deposited with the type of electrodes being tested. The cladding thickness shall be 3/16 inch minimum. The clad surfaces shall be machined or ground reasonably smooth before assembly of the test plate. When available, base plates having the same chemical composition as the deposited weld metal of the electrode being tested can be substituted for the clad plates.

4.5.2 Other tests. Table XII summarizes tests, in addition to those specified in table X required for qualification testing of the electrodes for the QPL and for quality conformance inspection, as applicable.

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TABLE X. Summary of required weld tests.[1/]

Weld type no.	Electrode type	Diameter sizes						Weld joint design procedure	Welding position[2/]	Tests		Chemical	Requirements
		3/32	1/8	5/32	3/16	7/32	1/4			X-ray[3/]	Tensile[4/]		
3	All	X	X	X	X	X	X	Figure 1 and notes	F	---	X	Chemical analysis[5/]	MIL-E-22200 and tables III and IV herein
8a	All	X	X	X	---	---	---	Figure 2 and notes	V	X	---	---	3.7.1 of MIL-E-22200
		---	---	---	X	X	X		F	X	---	---	

1/ For qualification testing of the electrodes for the QPL, all electrode types shall be tested using direct current (d.c.) reverse polarity (r.p.). In addition, MIL-XX16 and MIL-XX18 electrode types shall be tested using alternating current (a.c). For comparison inspection, when required, and quality conformance inspection, all electrode types shall be tested using d.c., r.p.

2/ V = vertical, F = flat.

3/ When both tests (3 and 8a) are applicable, radiography of test No. 3 will satisfy the requirements for test No. 8a for class 2 electrodes. If test No. 3 is radiographed, test No. 8a may be eliminated for class 2 electrodes.

4/ Includes ultimate, yield, elongation, reduction in area.

5/ May be determined from milling sample taken from ruptured ends of tensile specimens or drillings. In event of variances with composition requirements or conflicting results by testing activities, analysis shall be verified by procedures specified in 4.6.8 of MIL-E-22200.

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TABLE XI. Preheat, interpass, and postheat treatment.

Type	Temperature °F		
	Minimum preheat and interpass thickness (inches)		Postheat 1/ 2/ ± 25°F
	3/4 and under	Over 3/4	
MIL-80XX-B2 MIL-80XX-B2L	300	400	1300
MIL-90XX-B3 MIL-90XX-B3L	300	400	1350
MIL-502-XX MIL-502-XX-L	400	500	1375
MIL-505-XX MIL-410-XX	400 400	500 500	1375 1375

1/ For types MIL-80XX-B2L, MIL-90XX-B3L and MIL-502-XX-L, required only for qualification testing of these electrodes for the QPL (see notes to table III).

2/ For postheat treatment, the holding time at temperature shall be 1 hour per inch of thickness. For thicknesses less than 1 inch the holding time shall be proportional to the thickness but not less than 30 minutes. The maximum rate of raising or lowering the temperature shall be 200°F per hour.

TABLE XII. Summary of other tests.

Tests	Procedures	Requirements
	MIL-E-22200	
Visual and dimensional	4.6.2.3	3.10 (MIL-E-22200) and 3.6 herein
Concentricity	4.6.2	3.5.2 (MIL-E-22200)
Dielectric strength 1/	4.6.3	3.5.4 (MIL-E-22200)
Covering, flaking and cracking 1/	4.6.4	3.5.5 (MIL-E-22200)
Covering moisture	4.6.9	3.2.2.2 herein
Total iron content in covering 1/	4.6.10	3.2.2.1 herein
Alloy identity	4.6.11	3.6.4 (MIL-E-22200)

1/ Required for qualification (QPL) inspection only.

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4.6 Alloy identity certification. When specified (see 6.2), the Government representative shall affix his stamp and date on the container, containing electrodes conforming to 3.6.5 of MIL-E-22200. The letters "AT" shall be affixed to the container adjacent to the date to indicate "Alloy Tested".

4.7 Certificate of test. A certificate of quality conformance of the weld tests specified in 4.4.3 (i.e., x-ray, mechanical, and chemical analysis), containing actual quantitative test and inspection results including the range of amperage employed in making the required welds for quality conformance inspection for each accepted lot, heat, and wet mix, as applicable, included in a particular shipment shall be furnished to the consignee with the shipment. The certification shall include tests results from the quality conformance test and the wet mix equivalency test as applicable for the shipment. The minimum quality conformance test result data and wet mix equivalency test result data required shall be as shown on figures 3 and 4 and shall be submitted on the forms shown.

4.8 Packaging inspection. Packaging inspection shall be in accordance with 4.4.7 of MIL-E-22200.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2.1).

5.1.1 Level A. All electrodes shall be packaged in class 1 unit containers in accordance with MIL-W-10430 except when specified (see 6.2.1). Twelve-, 14-, and 18-inch electrodes shall be packaged in class 1b (50-pound) unit containers.

5.1.1.1 Special instructions. When electrodes are required to be packaged in class 1a unit containers in accordance with MIL-W-10430, fiberboard or other cushioning material shall be inserted in each end of the unit container to restrict the end-wise movement of the electrodes. The cushioning materials shall be dry and in equilibrium with the electrode covering. Class 1a containers shall be of a snug fitting design with the container length not to exceed the rod length by 1/2-inch.

5.1.2 Level C. Packaging shall afford protection against deterioration and physical damage during handling and shipment from the supply source to the using activity. The packaging applied shall insure conformance to the stability requirements specified in 3.5.8 of MIL-E-22200.

5.2 Packing. Packing shall be levels A, B, or C as specified (see 6.2.1).

5.2.1 Levels A, B, and C. Electrodes shall be packed level A, B, or C as specified (see 6.2.1) and where applicable palletized in accordance with MIL-W-10430.

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5.3 Marking. In addition to any special marking required herein or by the contract or order (see 6.2.1), interior packages and exterior shipping containers and palletized loads shall be marked in accordance with MIL-W-10430.

5.3.1 Special marking.

5.3.1.1 Interior packages and shipping containers. In addition to the lot identification required by 4.4.6 of MIL-E-22200, identification on interior packages shall include MIL type, size, specification number, manufacturer's or distributor's name, date of manufacture (month/year), and manufacturer's or distributor's brand or type designation. In addition, all unit packages shall be marked or labeled with the following precautionary information, or equal, as a minimum and prominently displayed:

WARNING: Protect yourself and others. Read and understand this label.

FUMES AND GASES can be dangerous to your health. ARC RAYS can injure eyes and burn skin. ELECTRIC SHOCK can kill.

Read and understand the manufacturer's instructions and your employer's safety practices.
Keep your head out of the fumes.
Use enough general ventilation or exhaust at the arc or both to keep fumes and gases from your breathing zone, and the general area.
Wear correct eye, ear and body protection.
Do not touch live electrical parts.
See American National Standard Z49.1 "Safety in Welding and Cutting" published by the American Welding Society, 2501 N.W. 7th St., Miami, FL 33125; OSHA Safety and Health Standards, 29 CFR 1910, are for sale from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

DO NOT REMOVE THIS LABEL

Significant toxic constituents when present in the electrode in greater than trace amounts shall be identified on the caution label so that normal ventilation can be increased accordingly. These constituents include but are not limited to those itemized in section 8.5 through 8.11 of AWS Z49.1.

5.3.1.2 All containers shall include the following marking:

"ASBESTOS FREE"

5.3.1.3 Class 1b containers. Each class 1b container on the container end adjacent to the electrode grip end shall be marked "OPEN THIS END".

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6. NOTES

6.1 Intended use. This specification is intended to cover several "low hydrogen" types of arc welding electrodes of various alloys for elevated temperature service and for corrosion resistance. The designation "low hydrogen" applies to electrodes having coverings which are substantially free from water and other compounds such as cellulose which will liberate hydrogen in the arc. Intended use of each type is as specified in 6.1.1 through 6.1.6.

6.1.1 Type MIL-80XX-B2 is intended for welding 1-1/4 percent chromium, 1/2 percent molybdenum steel in high temperature service up to 950°F.

6.1.2 Type MIL-90XX-B3 is intended for the welding of 2-1/4 percent chromium, 1 percent molybdenum steel in high temperature service up to 1050°F involving moderate conditions of corrosion or oxidation.

6.1.3 Type MIL-410-XX is intended for the welding of 12 percent chromium steel in service up to 1300°F where very good resistance to corrosion and oxidation is required. Where strength is a factor, it is seldom used above 1200°F.

6.1.4 Type MIL-502-XX is intended for the welding of 5 percent chromium, 1/2 percent molybdenum steel in high temperature service up to 1200°F involving severe corrosive or oxidizing conditions.

6.1.5 Type MIL-505-XX is intended for the welding of 9 percent chromium, 1-1/4 percent molybdenum steel in high temperature service up to 1300°F involving severe corrosive and oxidizing conditions. This alloy has good oxidation resistance up to 1500°F.

6.1.6 Low-carbon type. The L type electrode is intended for use in welding materials of the same composition that the respective higher carbon type is used. L types may be used where preheat and interpass temperature control is impractical, where the base material is 1 inch thick or less. The L types also may be used on heavier than 1 inch base material with preheat and interpass temperature control 100°F to 200°F below that required for the respective higher carbon types. The L types may also be used on base material less than 1 inch in thickness when post heat is impractical.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class and size required (see 1.2).
(c) For 3/32, 1/8, and 5/32 size electrodes, the length required (see table V).
(d) If rebaking is not required for quality conformance testing (see table VII, note 1).

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(e) Whether the alloy identity test and certification is required for the core wire of each finished electrode (see 4.6).
(f) Level of packaging, packing, and marking required (see 5.1, 5.2, and 5.3).

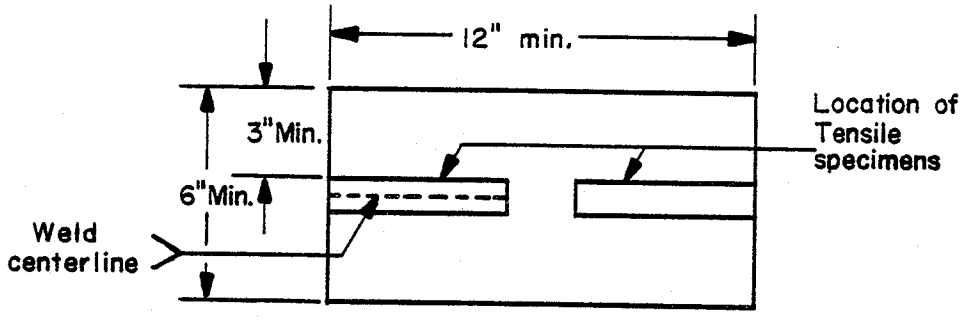
6.2.2 Electrodes should be ordered by the pound.

6.3 Inspection after delivery. Post delivery inspection of electrodes to determine conformity to this specification and for acceptance thereof is the responsibility of the consignee.

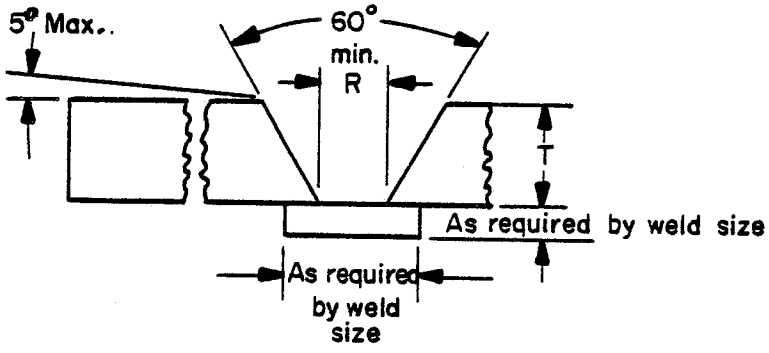
6.4 Changes from previous issue. Asterisks (*) are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
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Layout of test plate



SH 11729

FIGURE 1. Weld joint for all-weld-metal tension test (test No. 3).

Notes to figure 1:

1. Base metal shall be as specified in 4.5.1.2.
2. Types MIL-90XX-B3 and MIL-90XX-B3L shall be qualification tested and quality conformance tested after rebaking in accordance with the procedure outlined in note 1 of table VII unless specifically exempted by the acquisition document.

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Notes to figure 1 (Continued):

3. Welds shall be deposited in the flat position using d.c., r.p. and a.c. as required by note 1 to table X. The actual amperage values used shall be within the manufacturer's recommended current range and shall be recorded and reported. The root weld layer may be deposited with 1/8-inch diameter electrodes when testing 3/16-inch and larger diameter electrodes, provided both electrodes have the same manufacturer's designation. Welds shall be deposited in approximately 1/8-inch thick layers using a 2-1/2 core wire diameter maximum weave buildup sequence. The final layer shall include a reinforcement of standard proportions (1/32-inch to 1/8-inch). Each layer shall be started at the finishing end of the preceding layer. The operator error, pass and layer sequence shall be recorded and reported, as specified in table XIII.
4. After the joint has been tacked, the assembly shall be preheated as required by table XI. After completion of each layer the assembly shall be allowed to cool in still air at room temperature and the subsequent layer shall be started where the temperature of the plate has dropped to the interpass temperature of table XI. If it is necessary to interrupt this procedure, the assembly shall be slow cooled and preheated again.
5. Radiographic examination of test No. 3 is not required, but may be used in lieu of radiography of test No. 8a for class 2 electrodes when test No. 3 is radiographed. When used, the weld reinforcement and the backing shall be removed flush with the base plate surfaces. The assembly shall be radiographed in accordance with MIL-STD-271. Radiographs shall meet acceptance standards of NAVSEA 0900-LP-003-9000 as specified in 3.7.1 of MIL-E-22200.
6. The test weldment shall then be cut into tensile coupons as required by the notes to table III. No base metal shall be removed within 1/2-inch of the edge of the face of the weld by flame cutting. Only sawing or machining shall be used within 1/2-inch of the weld.
7. Coupons shall be stress-relieved when required in an electric furnace and in accordance with table XI and notes thereto, except when the whole test weldment was previously stress-relieved (see note 5 above).
8. The as-welded and stress-relieved tensile coupons shall be machined into one all-weld-metal tensile specimen for each test required by table III and notes thereto. The specimens shall be taken from the locations detailed on figure 1. The stress-relieved tensile coupon shall be etched, when necessary, to establish the location of the weld. The 0.505-inch diameter specimens shall be machined from 3/4-inch thick coupons and 0.357-inch diameter specimens from 1/2-inch thick coupons. See AWS B4.0 or ASTM A 370 for details. The specimens shall be tested for conformance with table III.

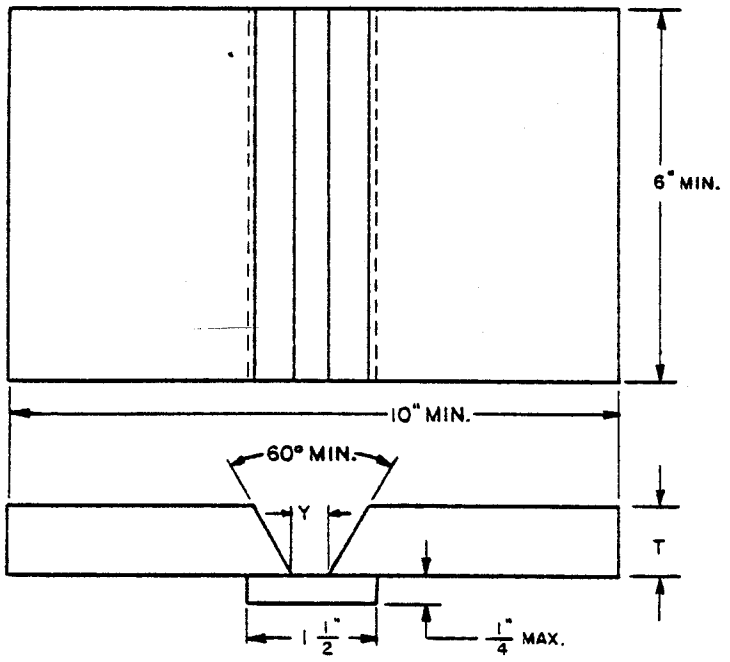
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TABLE XIII. Groove weld preparation.

Electrode size	Minimum thickness (T)	Minimum root opening (R)	Number of layers	
			Minimum	Maximum
(Inch)	(Inch)	(Inch)		
3/32	1/2	1/4	1/	1/
1/8	1/2	1/4	1/	1/
5/32	3/4	1/2	7	9
3/16	3/4	1/2	7	9
7/32	3/4	1/2	7	8
1/4	3/4	1/2	9	11

1/ Number of layers shall be reported.

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Electrode size	3/32 (Inch)	1/8 (Inch)	5/32 (Inch)	3/16-5/16 (Inch)
Root opening (Y)1/	1/4	5/16	3/8	1/2
Thickness (T)2/	1/4	3/8	1/2	1/2
Weld bead Width (max.)	3/8	1/2	5/8	3/4

1/ Tolerance ±1/16-inch.
2/ Minimum.

FIGURE 2. Groove weld usability test (test No. 8a).

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Notes to figure 2:

1. Welding shall be accomplished with the test plate in the vertical or flat position as specified in table X. Direct current, reverse polarity and alternating current shall be used in accordance with note 1 to table X. The actual amperage values used shall be within the manufacturer's recommended current range and shall be recorded and reported.
2. Base metal shall be in accordance with 4.5.1.2 herein.
3. The edge preparation shall be flame cut and ground to remove slag and surface irregularities.
4. Welds shall be deposited in approximately 1/8-inch thick layers. Weld bead widths shall be as shown above. Each bead shall contain a start in the area to be evaluated.
5. The reinforcement shall be 3/16-inch maximum.
6. Preheat and interpass temperature shall be as specified in table XI.
7. After completion of the weld, reinforcement and backing strips shall be removed flush with the base plate on both surfaces and the assembly shall be x-rayed in accordance with MIL-STD-271.
8. The radiograph shall meet the x-ray standards of NAVSEA 0900-LP-003-9000 as specified in 3.7.1 of MIL-E-22200.

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CERTIFICATION OF QUALITY CONFORMANCE TESTS

Manufacturer or Distributor _____ Customer's Name _____

Address _____

Date _____ Customer's Order No. _____

Specification MIL-_____

Type MIL-_____ Core Wire Heat No. _____

Diameter & Length _____ Lot Identification MIL-E-22200 Para. _____

Inspection Level _____

Lot No. _____ Wet mix No. _____

Chemical analysis (Complete)

Carbon _____

Manganese _____

Phosphorus _____

Sulfur _____

Silicon _____

Chromium _____

Molybdenum _____

Mechanical test

	As-welded	Stress-relieved
Yield strength (0.2% offset method)	_____	_____
Tensile strength	_____	_____
Elongation (%)	_____	_____
Reduction of area (%)	_____	_____

Chemistry was taken from: ____ Chem Pad ____ Groove Weld

X-Ray Results _____

Concentricity (%) _____

Covering Moisture _____

Covering Iron Content _____

Groove Weld Test

Test No.	3	8a	Chem Pad
Amperage	___	___	_____

Operator Error (Layer Nos.) _____

FIGURE 3. Certification of quality conformance tests.

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Grinding during 8a test plate preparation (see note 3)

Alloy identity tests:

(a) Core wire _____

(b) Finished electrode _____ (if required, see 3.7)

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____

NOTES:

1. Items not applicable to the type electrode involved or exempted by the acquisition document shall be marked N/A.
2. This form is applicable to the quality conformance tests required for each lot. Figure 4 forms shall be employed to report and certify the wet mix equivalency tests where applicable for each wet mix of covering mixture in the lot. One copy of the figure 4 form shall be employed for certifying equivalency of each wet mix in a lot where applicable. One copy of the quality conformance certificate (figure 3) for the lot shall be forwarded with copies of the mix equivalency certificates for each additional mix from that lot that is included in the shipment.
3. See 3.5.7 and 3.9 of MIL-E-22200.

FIGURE 3. Certificate of quality conformance tests. - Continued

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CERTIFICATION OF WET MIX EQUIVALENCY TESTS

Manufacturer or Distributor _____ Customer's Name _____

Address _____

Date _____ Customer's Order No. _____

Specification MIL-_____

Type MIL-_____

Diameter Size _____ Core Wire Heat No. _____

Lot No. _____ Wet Mix No. _____

Sampling
Chemical Analysis

Manganese _____

Silicon _____

Chromium _____

Molybdenum _____

X-Ray Results _____

Chemistry was taken from:

Chem Pad Groove Weld

Test No. _____

	Chem Pad	8a
Amperage	_____	_____

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____

NOTES:

1. Items not applicable to the type electrode involved or exempted by the acquisition document shall be marked N/A.
2. One copy of the figure 4 form shall be employed for certifying equivalency of each wet mix in a lot where applicable. One copy of the quality conformance certificate (figure 3) for the lot shall be forwarded with copies of the mix equivalency certificates for each additonal mix from that lot that is included in the shipment.

FIGURE 4. Certification of wet mix equivalency tests.

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